

PROJECT

AIM:

Marksheet Generator Tool :

The Marksheet Generator Tool is a comprehensive software that utilizes Object-Oriented Design to efficiently calculate total marks, compute percentages, and generate grades based on performance. It allows for flexible subject addition, ensuring adaptability to various academic curricula. With its easy-to-read marksheet output, users can effortlessly track student progress.

Key benefits include:

- Calculating total marks and percentages
- Assigning grades based on predefined criteria (A+ to Fail)
- Generating comprehensive marksheets for easy academic Tracking.

Project Features:

- Object-Oriented Design
- Calculates total marks
- Computes percentage
- Generates grade based on performance
- Flexible subject addition
- Easy-to-read marksheet output

Key Functionalities:

1. Add multiple subject marks
2. Calculate total marks
3. Compute percentage
4. Assign grades
5. Generate comprehensive marksheet

Grading Criteria:

- A+: 90% and above
- A: 80-89%
- B+: 70-79%
- B: 60-69%
- C: 50-59%
- Fail: Below 50%

CODE:

```
print("\n*** Class 10th Student Marksheet ***\n")

# Input details
Name = input("Enter your Name: ")
Roll_no = int(input("Enter your Roll No: "))
Roll_code = int(input("Enter your Roll Code: "))
school_name=input("Enter school name: ")

# Input marks
sub1 = int(input("Enter the marks for Mathematics: "))
sub2 = int(input("Enter the marks for Hindi: "))
sub3 = int(input("Enter the marks for Science: "))
sub4 = int(input("Enter the marks for Social Science: "))
sub5 = int(input("Enter the marks for Sanskrit: "))

# Calculate total marks and percentage
total = sub1 + sub2 + sub3 + sub4 + sub5
percentage = (total / 500) * 100

graced = False # if grace marks were given

# Apply Grace Marks (if needed)
subjects = [sub1, sub2, sub3, sub4, sub5]

for i in range(len(subjects)):
    if subjects[i] < 50 and subjects[i] >= 45: # add by at most 5 marks
        print(f"Adding {50 - subjects[i]} grace marks to Subject {i+1}")
        total =total+ (50- subjects[i])
        subjects[i] = 50
        graced = True

# Recalculate percentage after grace marks
percentage = (total / 500) * 100
```

```
# Display marksheet
print("\n***** Marksheet *****")
print(f"Mathematics: {sub1}")
print(f"Hindi: {sub2}")
print(f"Science: {sub3}")
print(f"Social Science: {sub4}")
print(f"Sanskrit: {sub5}")
print("-----")
print(f"Total Marks: {total} / 500")
print(f"Percentage: {percentage:.2f}%")

# Determine Pass/Fail status
if percentage >= 90:
    print("Result: A+")
elif percentage >= 80:
    print("Result: A")
elif percentage >= 70:
    print("Result: B+")
elif percentage >= 60:
    print("Result: B")
elif percentage >= 50:
    print("Result: C")
elif graded:
    print("Result: Pass (With Grace Marks)")
else:
    print("Result: Fail")

print("*****")
```

OUTPUT:

```
*** Class 10th Student Marksheet ***
```

```
Enter your Name: PIYUSH YADAV
Enter your Roll No: 220045
Enter your Roll Code: 22005
Enter school name: MAHATMA GANDHI SMARAK +2 HIGH SCHOOL, JHAJHA, JAMUI
Enter the marks for Mathematics: 90
Enter the marks for Hindi: 89
Enter the marks for Science: 95
Enter the marks for Social Science: 86
Enter the marks for Sanskrit: 97
```

```
***** Marksheet *****
```

```
Mathematics: 90
Hindi: 89
Science: 95
Social Science: 86
Sanskrit: 97
```

```
-----
Total Marks: 457 / 500
Percentage: 91.40%
Result: A+
```

```
*****
```

```
*** Class 10th Student Marksheet ***
```

```
Enter your Name: PAWAN YADAV
Enter your Roll No: 220242
Enter your Roll Code: 22504
Enter school name: MAHATMA GANDHI +2 SAMARAK HIGH SCHOOL, JHAJHA, JAMUI
Enter the marks for Mathematics: 50
Enter the marks for Hindi: 50
Enter the marks for Science: 50
Enter the marks for Social Science: 50
Enter the marks for Sanskrit: 49
Adding 1 grace marks to Subject 5
```

```
***** Marksheet *****
```

```
Mathematics: 50
Hindi: 50
Science: 50
Social Science: 50
Sanskrit: 49
```

```
-----
Total Marks: 250 / 500
Percentage: 50.00%
Result: C
```

```
*****
```